

Day 2 – Operators

1. Check if number lies in range (logical)
2. Bitwise AND, OR, XOR demo
3. Implement a calculator with all operators
4. Bitwise operations: swap two numbers
5. Bitwise operations: Find odd or even
6. Check if two numbers are co-prime
7. Find the HCF of the number

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1. Check if number lies in range (logical)

```
num = int(input("Enter number: "))
start = int(input("Enter start range: "))
end = int(input("Enter end range: "))
if num >= start and num <= end:
    print("Number is in range")
else:
    print("Number is not in range")
```

2. Bitwise AND, OR, XOR demo

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("AND =", a & b)
print("OR =", a | b)
print("XOR =", a ^ b)
```

3. Implement a calculator with all operators

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("1.Add")
print("2.Subtract")
print("3.Multiply")
print("4.Divide")
print("5.Modulus")x
choice = int(input("Enter choice: "))
```

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```
if choice == 1:
    print("Result =", a + b)
elif choice == 2:
    print("Result =", a - b)
elif choice == 3:
    print("Result =", a * b)
elif choice == 4:
    print("Result =", a / b)
elif choice == 5:
    print("Result =", a % b)
else:
    print("Invalid Choice")
```

4. Bitwise operations: swap two numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
a = a ^ b
b = a ^ b
a = a ^ b
print("a =", a)
print("b =", b)
```

5. Bitwise operations: Odd or Even

```
n = int(input("Enter a number: "))
if n & 1:
    print("Odd")
else:
    print("Even")
```

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6. Check if Two Numbers are Co-Prime

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
small = a
if b < a:
    small = b
gcd = 1
for i in range(1, small + 1):
    if a % i == 0 and b % i == 0:
        gcd = i
if gcd == 1:
    print("Co-Prime Numbers")
else:
    print("Not Co-Prime Numbers")
```